# Network

**Network** is a group of computers connected with each other through wires, optical fibres or optical links so that various devices can interact with each other through a network.

## Networking

Computer networking refers to interconnected computing devices that can exchange data and share resources with each other.

## Overview of Data Communication Networks

Communication is defined as a process in which more than one computer transfers information, instructions to each other and for sharing resources. Or in other words, communication is a process or act in which we can send or receive data.

The effectiveness of any data communications system depends upon the following **four** fundamental characteristics:

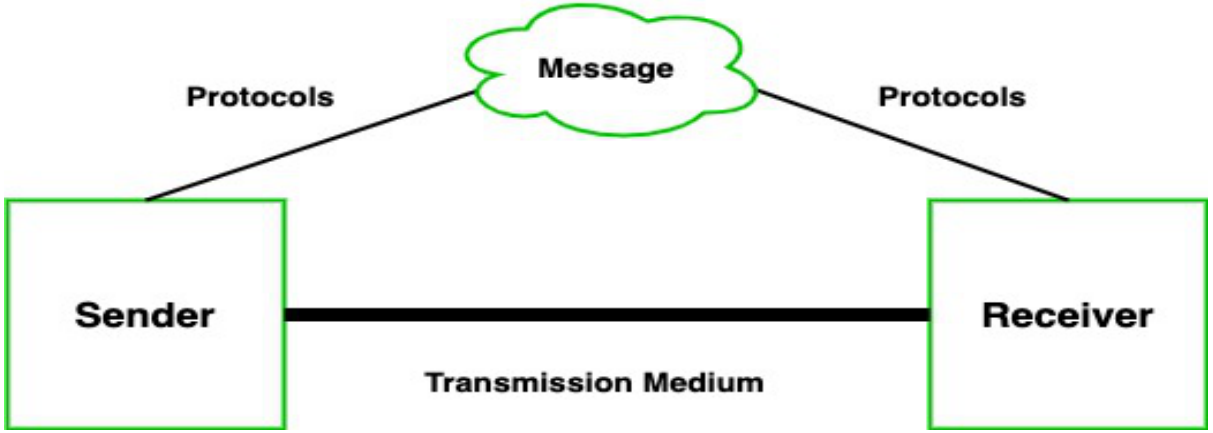
# Components of Data Communication

**Message:** A message is a piece of information that is to be transmitted from one person to another. It could be a text file, an audio file, a video file, etc.

**Sender:** It is simply a device that sends data messages. It can be a computer, mobile, telephone, laptop, video camera, or workstation, etc.
Receiver: It is a device that receives messages. It can be a computer, telephone mobile, workstation, etc.

**Transmission Medium / Communication Channels**: Communication channels are the medium that connect two or more workstations. Workstations can be connected by either wired media or wireless media.

**Set of rules (Protocol):** When someone sends the data (The sender), it should be understandable to the receiver also otherwise it is meaningless. For example, Sonali sends a message to Chetan. If Sonali writes in Hindi and Chetan cannot understand Hindi, it is a meaningless conversation.



# Type of data communication

**The data communication is divided into three types:**

1. **Simplex Communication**: It is one-way communication or we can say that unidirectional communication in which one device only receives and another device only sends data and devices uses their entire capacity in transmission. For example, IoT, entering data using a keyboard, listing music using a speaker, etc.

2. **Half Duplex communication**: It is a two-way communication, or we can say that it is a bidirectional communication in which both the devices can send and receive data but not at the same time. When one device is sending data then another device is only receiving and vice-versa. For example, walkie-talkie.

3. **Full-duplex communication:** It is a two-way communication or we can say that it is a bidirectional communication in which both the devices can send and receive data at the same time. For example, mobile phones, landlines, etc.

# Protocol and Standard in Computer Networks

Protocols and standards are important in computer networks. They are like the rules and guidelines that allow different devices and systems to communicate and work together smoothly. Protocols define how data is sent, received, and processed, while standards ensure that various technologies are compatible with each other.

Network protocol ensures that different technologies and components of the network are compatible with one another, reliable, and able to function together.

## What is Protocol?

A protocol is a set of rules that determines how data is sent and received over a network. The protocol is just like a language that computers use to talk to each other, ensuring they understand and can respond to each other's messages correctly. Protocols help make sure that data moves smoothly and securely between devices on a network.

In the above diagrams, Protocols are shown as a set of rules. Communication between the Sender and Receiver is not possible without Protocol.

# Key Elements of Protocol

- **Timing:** Timing refers to the synchronization and coordination between devices while transferring the data. Timing ensures at what time data should be sent and how fast data can be sent. For example, If a sender sends 100 Mbps but the receiver can only handle 1 Mbps, the receiver will overflow and lose data. Timing ensures preventing of data loss, collisions, and other timing-related issues.

- **Sequence Control**: Sequence control ensures the proper ordering of data packets. The main responsibility of sequence control is to acknowledge the data while it get received, and the retransmission of lost data. Through this mechanism, the data is delivered in correct order.

• **Flow Control:** Flow control regulates device data delivery. It limits the sender's data or asks the receiver if it's ready for more. Flow control prevents data congestion and loss.

• **Error Control:** Error control mechanisms detect and fix data transmission faults. They include error detection codes, data resend, and error recovery. Error control detects and corrects noise, interference, and other problems to maintain data integrity.

• **Security :** Network security protects data confidentiality, integrity, and authenticity. which includes encryption, authentication, access control, and other security procedures. Network communication's privacy and trustworthiness are protected by security standards.

# Types of Protocol

Network Layer Protocols

Transport Layer Protocols

Application Layer Protocol

Wireless Protocols

Routing Protocols

Security Protocols

Internet Protocols

# Important Protocols Used in Computer Network

**TCP (Transmission Control Protocol):** Ensures data is sent and received accurately by breaking it into packets, sending them, and reassembling them at the destination.

**IP (Internet Protocol):** Addresses and routes the packets to make sure they reach the right destination.

**HTTP/HTTPS (Hypertext Transfer Protocol/Secure)** HTTP used for transferring web pages on the internet.

**FTP (File Transfer Protocol)** Used for transferring files between computers on a network. It allows users to upload and download files.

**SMTP (Simple Mail Transfer Protocol):** Used for sending emails. It transfers emails from a client to a server or between servers.

**DNS (Domain Name System):** **DNS:** DNS stands for Domain Name System. An IP address is used to identify the connection of a host to the internet uniquely. But, people prefer to use the names instead of addresses. Therefore, the system that maps the name to the address is known as Domain Name System.

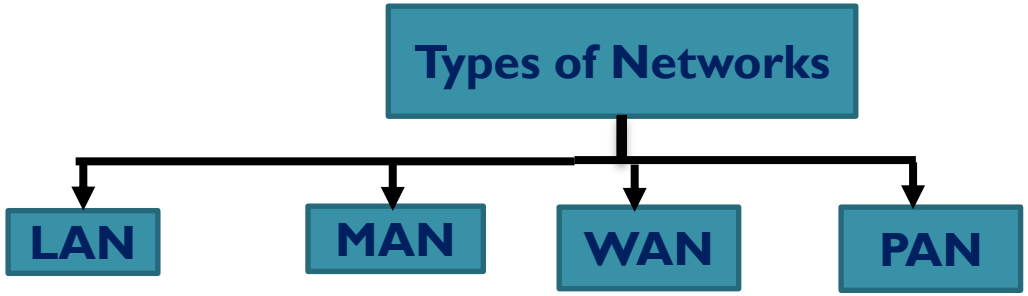
**DHCP (Dynamic Host Configuration Protocol):** Automatically assigns IP addresses to devices on a network, ensuring each device has a unique address.

**SSH (Secure Shell):** Provides a secure way to access and manage devices over a network. It encrypts the data, making it safe from eavesdropping.

**SNMP (Simple Network Management Protocol):** Used for managing and monitoring network devices like routers, switches, and servers. It collects and organizes information about these devices

# Types of Networks

- Local Area Network (LAN)
- Wide Area Network (WAN)
- Metropolitan Area Network (MAN)
- Personal Area Network (PAN)



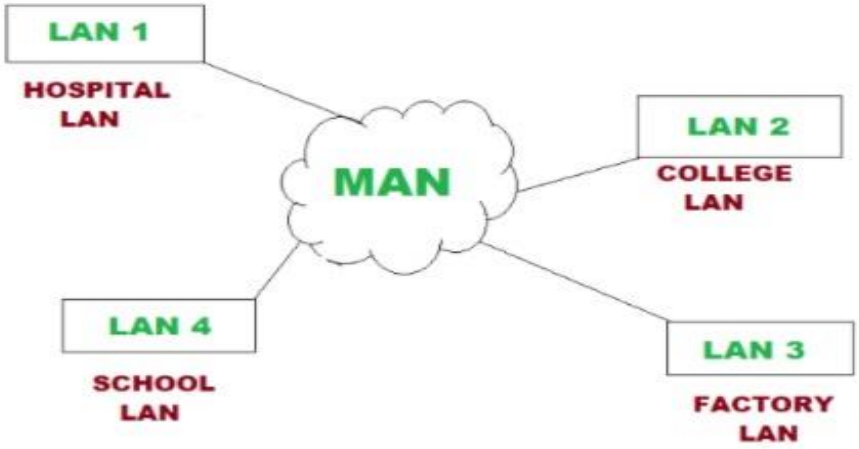

## LAN (Local Area Network)

- Local Area Network is a group of computers connected to each other in a small area such as building, office.
- LAN is used for connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.
- It is less costly as it is built with inexpensive hardware such as hubs, network adapters, and Ethernet cables.

- **The data is transferred at an extremely faster rate in Local Area Network.**

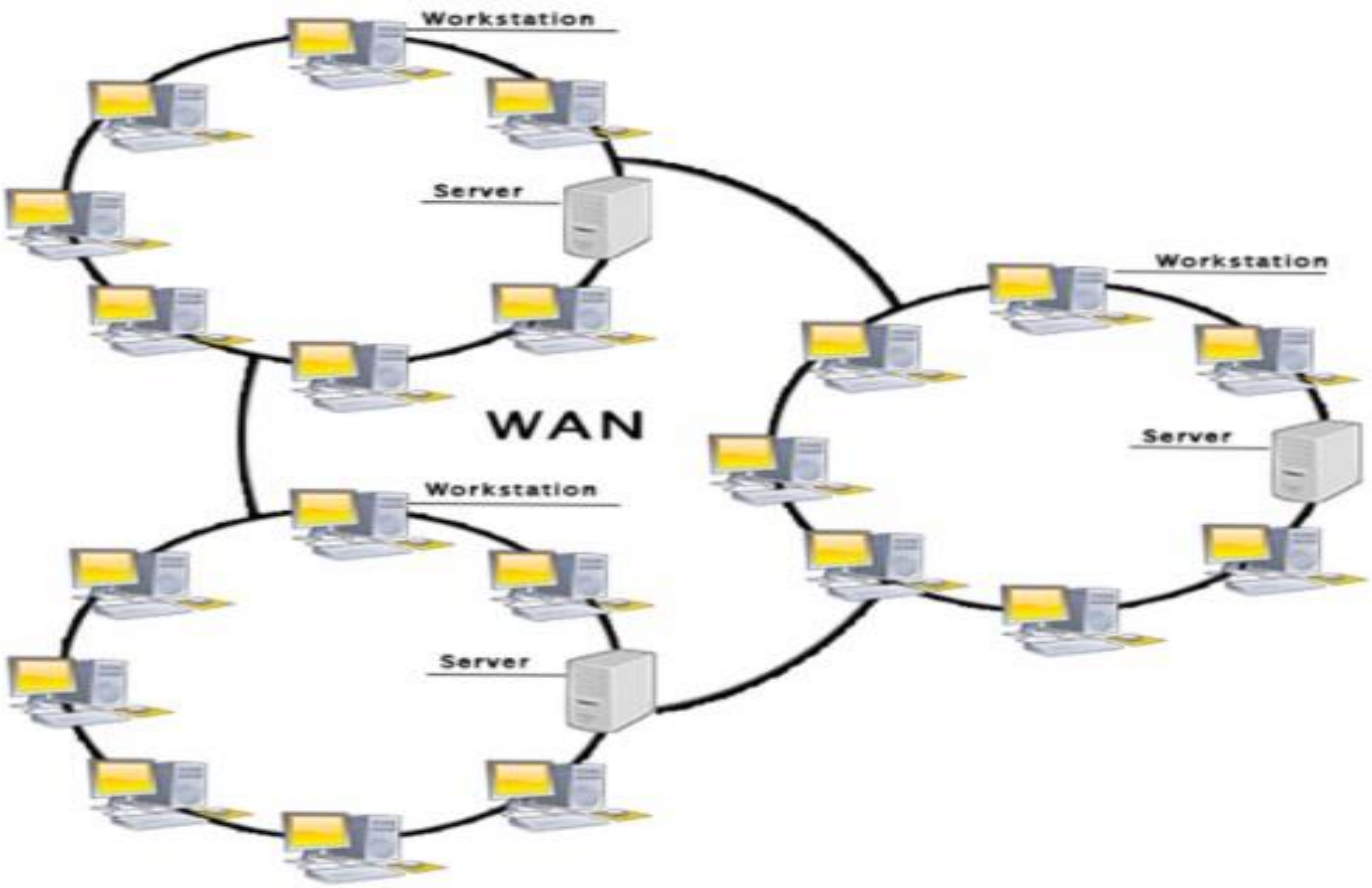
## Metropolitan Area Network (MAN)

**MAN or Metropolitan area Network covers a larger area than that of a LAN and smaller area as compared to WAN. It connects two or more computers that are apart but resides in the same or different cities. It covers a large geographical area and may serve as an ISP (Internet Service Provider).**




## WAN(Wide Area Network)

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- A Wide Area Network is quite bigger network than the LAN & MAN.
- A Wide Area Network is not limited to a single location, but it spans over a large geographical area through a telephone line, fibre optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- A Wide Area Network is widely used in the field of Business, government, and education.


Workstation
Server
Workstation
WAN
Server
Workstation
Server

**Examples Of Wide Area Network:**

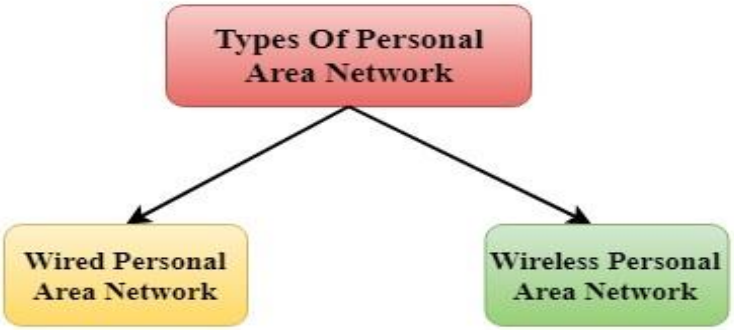
- **Mobile Broadband:** A 4G network is widely used across a region or country.
- **Last mile:** A telecom company is used to provide the internet services to the customers in hundreds of cities by connecting their home with fiber.
- **Private network:** A bank provides a private network that connects the 44 offices. This network is made by using the telephone leased line provided by the telecom company.

**PAN(Personal Area Network)**

- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.
- Personal Area Network is used for connecting the computer devices of personal use is known as Personal Area Network.
- **Thomas Zimmerman** was the first research scientist to bring the idea of the Personal Area Network.
- Personal Area Network covers an area of 30 feet.
- Personal computer devices that are used to develop the personal area network are the laptop, mobile phones, media player and play stations.

**There are two types of Personal Area Network:**

- **Wired Personal Area Network**
- **Wireless Personal Area Network**



**Wireless Personal Area Network: Wireless Personal Area Network is developed by simply using wireless technologies such as Wi-Fi, Bluetooth. It is a low range network.**

**Wired Personal Area Network: Wired Personal Area Network is created by using the USB.**

## Examples Of Personal Area Network:

- **Body Area Network:** Body Area Network is a network that moves with a person.

**For example**, *a m*obile network moves with a person. Suppose a person establishes a network connection and then creates a connection with another device to share the information.

- **Offline Network:** An offline network can be created inside the home, so it is also known as a **home network**. A home network is designed to integrate the devices such as printers, computer, television but they are not connected to the internet.
- **Small Home Office:** It is used to connect a variety of devices to the internet and to a corporate network using a *VPN.*